Tutorial 2, Question 3

Profit maximisation and the price markup

1. Profit maximisation

Profit = Return − Cost:

$$\pi(P, y) = P(y)\, y - C(y).$$

First-order condition with respect to output y:

$$\frac{d\pi}{dy} = P(y) + y\,\frac{dP(y)}{dy} - \frac{dC(y)}{dy} = 0$$

$$P + y\,\frac{dP}{dy} = \frac{dC}{dy} \qquad (MR = MC) \tag{①}$$

From ①:

$$P\left(1 + \frac{y}{P}\,\frac{dP}{dy}\right) = \frac{dC}{dy} \tag{②}$$

2. Price elasticity of demand

Denote the price elasticity of demand by η:

$$\eta = -\,\frac{dy/y}{dP/P} = -\,\frac{P}{y}\,\frac{dy}{dP} \quad \Longrightarrow \quad \frac{y}{P}\,\frac{dP}{dy} = -\frac{1}{\eta}.$$

Substituting η into ②:

$$P\left(1 - \frac{1}{\eta}\right) = MC = \frac{dC}{dy} \tag{③}$$

3. Labour as the only input

$$C = WN, \qquad y = f(N),$$

$$\frac{dC}{dN} = W, \qquad \frac{dy}{dN} = f_N = MPL.$$

$$\frac{dC}{dy} = \frac{dC}{dN} \cdot \frac{dN}{dy} = W \cdot \frac{1}{MPL} \tag{④}$$

Substituting ④ into ③:

$$\boxed{P\left(1 - \frac{1}{\eta}\right) = \frac{W}{MPL}}$$

4. The markup

$$\mu^c = \frac{P - MC}{MC} = \frac{P}{MC} - 1 \qquad ⑤$$

Notation. μ^c is the *markup on cost*: the percentage by which price exceeds marginal cost, measured relative to MC. The tutorial answer writes this as μ; we follow the textbook and write μ^c, because in Carlin & Soskice μ denotes the *markup on price*:

$$\mu^c = \frac{P - MC}{MC} = \frac{1}{\eta - 1} \quad \text{(on cost)}, \qquad \mu = \frac{P - MC}{P} = \frac{1}{\eta} \quad \text{(on price)}, \qquad \frac{1}{1 + \mu^c} = 1 - \mu.$$

E.g. $\eta = 5$: $\mu^c = 0.25$ (price is 25% above cost) but $\mu = 0.2$ (20% of the price is markup). The PS curve $w = \lambda(1 - \mu)$ uses the price markup μ.

From eq. ③:

$$1 - \frac{1}{\eta} = \frac{MC}{P} \quad \Longrightarrow \quad \frac{P}{MC} = \frac{1}{\frac{\eta - 1}{\eta}} = \frac{\eta}{\eta - 1}.$$

Substituting into ⑤:

$$\mu^c = \frac{\eta}{\eta - 1} - 1 = \frac{\eta - (\eta - 1)}{\eta - 1} \quad \Longrightarrow \quad \boxed{\mu^c = \frac{1}{\eta - 1}}$$

5. Price as a markup on marginal cost

Have

$$P\left(1 - \frac{1}{\eta}\right) = \frac{W}{MPL}.$$

Using $\mu^c = \frac{1}{\eta - 1}$:

$$1 + \mu^c = \frac{\eta - 1 + 1}{\eta - 1} = \frac{\eta}{\eta - 1}, \qquad \frac{1}{1 + \mu^c} = \frac{\eta - 1}{\eta} = 1 - \frac{1}{\eta}.$$

$$\Longrightarrow \quad P\left(\frac{1}{1 + \mu^c}\right) = \frac{W}{MPL} \quad \Longrightarrow \quad \boxed{P = (1 + \mu^c)\,\frac{W}{MPL}}$$

Intuition

Higher competition ⇒ $\eta \uparrow$ ⇒ lower μ^c.
Lower competition ⇒ lower η ⇒ higher μ^c.
Under perfect competition: $\eta \to \infty \Rightarrow \mu^c = 0$ (so $P = MC = W/MPL$).

6. Question 4: from price setting to price growth

Question: use the price-setting equation to show the relationship between wage growth and price growth. What if markups were not constant, but depended positively on inflation?

6.1 Constant markup

From Q3 we have the PS equation. Writing labour productivity as $\lambda = MPL$ (and keeping the textbook's μ^c; the tutorial answer writes μ):

$$P(t) = (1 + \mu^c)\,\frac{W(t)}{\lambda(t)} \qquad ⑥$$

Taking logs of ⑥ turns the product into a sum:

$$\ln P(t) = \ln(1+\mu^c) + \ln W(t) - \ln \lambda(t).$$

Totally differentiating with respect to time t:

$$\frac{d\ln P}{dt} = \frac{d\ln(1+\mu^c)}{dt} + \frac{d\ln W}{dt} - \frac{d\ln \lambda}{dt}.$$

By the chain rule, $\dfrac{d\ln x}{dt} = \dfrac{d\ln x}{dx}\cdot\dfrac{dx}{dt}$, and since $\dfrac{d\ln x}{dx} = \dfrac{1}{x}$, each term is a growth rate:

$$\frac{1}{P}\frac{dP}{dt} = \frac{1}{1+\mu^c}\frac{d(1+\mu^c)}{dt} + \frac{1}{W}\frac{dW}{dt} - \frac{1}{\lambda}\frac{d\lambda}{dt}. \qquad ⑦$$

In discrete time (period by period) this holds approximately:

$$\left(\frac{\Delta P}{P}\right)_t \approx \left(\frac{\Delta W}{W}\right)_t - \left(\frac{\Delta \lambda}{\lambda}\right)_t + \left(\frac{\Delta(1+\mu^c)}{1+\mu^c}\right)_t.$$

If the markup is constant, $\Delta(1+\mu^c) = 0$ and the last term drops out. This gives the **price growth equation**:

$$\boxed{\left(\frac{\Delta P}{P}\right)_t = \left(\frac{\Delta W}{W}\right)_t - \left(\frac{\Delta \lambda}{\lambda}\right)_t}$$

price inflation = wage inflation − productivity growth.

Firms protect their profit margin by passing every wage increase straight into prices. With constant productivity ($\Delta\lambda = 0$) as in the baseline model, wage inflation passes into price inflation one-for-one, and the real wage $W/P = \lambda/(1+\mu^c)$ stays constant.

6.2 Markup rising with inflation

Assumption.

$$\mu_t^c = \mu^c(\pi_t), \qquad \mu^{c\prime}(\pi) > 0$$
$$P_t = (1+\mu_t^c)\frac{W_t}{\lambda_t}$$

Step 1: price growth. The markup term in ⑦ no longer drops out:

$$\pi_t \approx \omega_t - g_{\lambda,t} + \Delta\ln(1+\mu_t^c) \qquad ⑧$$

where $\pi_t = \left(\dfrac{\Delta P}{P}\right)_t$, $\omega_t = \left(\dfrac{\Delta W}{W}\right)_t$, $g_{\lambda,t} = \left(\dfrac{\Delta\lambda}{\lambda}\right)_t$.

Step 2: the markup term. To first order,

$$\Delta\ln(1+\mu_t^c) \approx k\,\Delta\pi_t, \qquad k \equiv \frac{\mu^{c\prime}(\pi)}{1+\mu^c} > 0.$$

When inflation is rising ($\Delta\pi_t > 0$), this term is positive.

Step 3: solve for π_t. Substituting into ⑧:

$$\pi_t \approx \omega_t - g_{\lambda,t} + k(\pi_t - \pi_{t-1})$$

$$\Longrightarrow \quad \boxed{\pi_t \approx \frac{1}{1-k}\left(\omega_t - g_{\lambda,t}\right) - \frac{k}{1-k}\pi_{t-1}} \qquad (0 < k < 1)$$

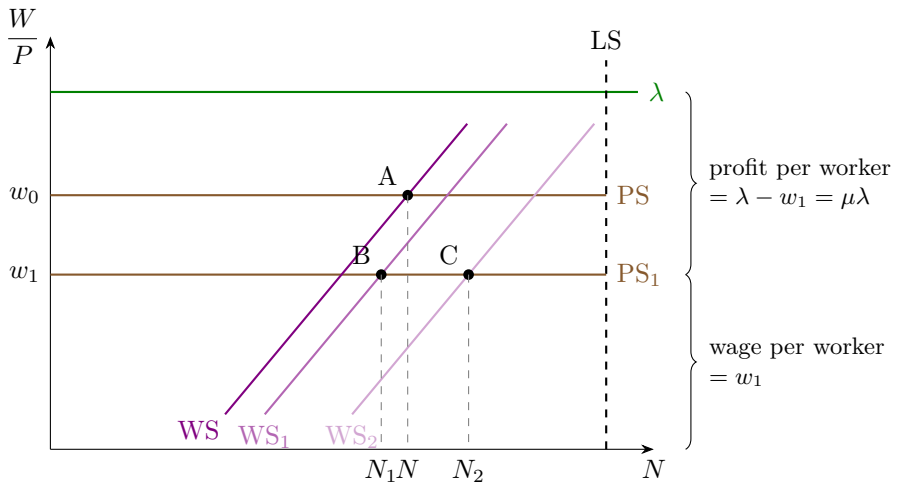
Result. Since $\dfrac{1}{1-k} > 1$, wage inflation passes into price inflation *more than one-for-one*: price inflation is **amplified** relative to the constant-markup case.

7. Question 5: firm power, union strength and inequality

Question: consider an increase in the average market power of firms combined with a fall in union strength. Use the WS-PS and Lorenz curve diagrams to illustrate the expected change in inequality.

Notation. In this question μ is the textbook's *markup on price*, so the PS curve is $w^{PS} = \lambda(1-\mu)$ and μ is the profit share. Higher market power raises μ^c and μ together.

7.1 WS-PS diagram



- **Increase in firm power** ⇒ lower competition ⇒ higher μ ⇒ PS shifts down to PS_1.
- **Fall in union strength** ⇒ wage-setting power ↓ ⇒ WS shifts right, to WS_1 or WS_2.

Results.

(a) The real wage falls unambiguously: $w_0 \to w_1$, since $w = \lambda(1-\mu)$ and $\mu \uparrow$.

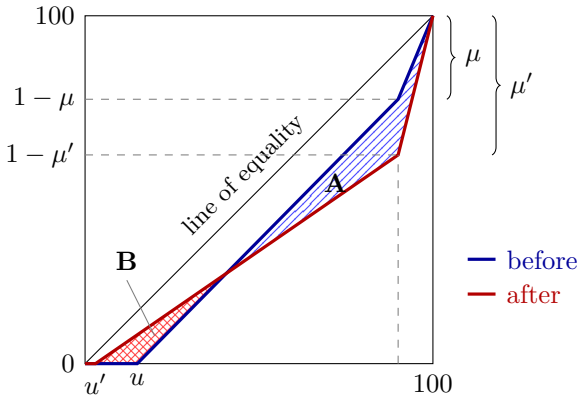
(b) The effect on employment is ambiguous and depends on the relative size of the two shocks:

- WS shifts only to WS_1 (point B): $N_1 < N$, employment falls.
- WS shifts further to WS_2 (point C): $N_2 > N$, employment rises.

7.2 Lorenz curve

The Lorenz curve has three groups: the unemployed (income 0), workers (wage share $1-\mu$) and owners (profit share μ). The shock affects it through two channels:

- **Distribution channel:** $\mu' > \mu$, so the kink moves down and the curve moves away from the line of equality (area A, +).
- **Unemployment channel:** the share of the population with zero income changes.



- **Unemployment rises** (WS_1, $u' > u$): both channels push the curve away from the line of equality, so inequality *rises unambiguously* (as in textbook Figure 2.19).
- **Unemployment falls** (WS_2, $u' < u$, as drawn): fewer people have zero income, so the bottom of the curve moves towards the line of equality (area B, $-$). Since Gini $= 2 \times$ (area between equality line and Lorenz curve), $\Delta\text{Gini} = 2(A - B)$:

$$A > B: \text{ inequality rises;} \qquad A < B: \text{ inequality falls;} \qquad A = B: \text{ no change.}$$